

MarketMind AI: Quantitative Equity Platform & Autonomous Voice Copilot

An institutional-grade stock intelligence platform designed for 270+ National Stock Exchange (NSE) equities, researched with AI assistance to replace retail emotional guessing with real-time order book imbalances, causal supply chain ripple effects, and hands-free voice control.

The Retail Problem (Why 90% Lose Money)

According to SEBI data, over 90% of retail traders lose capital. They trade based on delayed news, WhatsApp rumors, and lagging indicators (RSI/MACD). When a trade goes into loss, they hold onto it with "hope" until wiped out.

The AI Institutional Solution

MarketMind AI operates like institutional quant hedge funds. It measures live buyer vs. seller queue depth, true institutional average prices, economic domino reactions, and mathematically enforced stop-loss exit points.

The 5 Core Intelligence Engines (Explained Simply — No Formulas)

FASTAPI BACKEND

1. Directional Probability Model (recommendations_service.py)

BUYER VS. SELLER STRENGTH

What It Does: Calculates the exact percentage chance (e.g. 66% Upward Probability) that a stock will continue rising today, instead of giving a subjective guess.

How It Works: Evaluates 3 real-world clues: (1) Are there more active limit buyers than sellers in the exchange order queue? (2) Is the price trading above today's institutional average price (VWAP)? (3) Is trading volume unusually heavy? It turns these 3 signals into a clear 0% to 100% confidence score.

Dalal Street Example: Syрма SGS has huge buyer demand at ₹1,635 with double normal volume. The engine computes a 66.5% probability of upward continuation and issues a *STRONG BUY* signal targeting ₹1,729.00.

2. Macro Domino Shock Engine (domino_service.py)

SUPPLY CHAIN RIPPLE EFFECTS

What It Does: Tracks how a change in global commodities (Crude Oil, Gold) or RBI interest rates cascades step-by-step through company supply chains and profit margins.

How It Works: When raw material prices spike, it automatically traces who suffers higher costs and who makes higher profits across connected corporate sectors.

Dalal Street Example: If Crude Oil jumps +20%, IndiGo Airlines loses profits because aviation fuel becomes expensive, while ONGC gains +₹1,120 Crore because it drills and sells crude oil at higher prices.

3. Stock DNA Behavioral Twin Finder (dna_service.py)

SMART STOCK MATCHMAKER

What It Does: Finds "twin" stocks that share the same financial DNA (growth, valuation, debt, and momentum) so you never have to chase an overpriced stock.

How It Works: Analyzes 8 fundamental and technical characteristics across 270 NSE companies to detect stocks that behave almost identically in the market.

Dalal Street Example: When Tata Motors rallies +35% and becomes too risky to buy, Stock DNA scans 270 stocks and points you to Bharat Forge (89% twin match), allowing you to capture the auto boom at a safer price.

4. 1-Day Monte Carlo Risk Gauge

PORTFOLIO_SERVICE.PY

What It Does: Simulates 1,000 possible market days to tell you the maximum realistic loss on your portfolio in plain Rupees.

How It Works: Uses 1 year of real historical price swings to stress-test your holdings across 1,000 randomized scenarios.

Example: For a ₹10 Lakh portfolio in Reliance and Adani, it certifies: "On 95 out of 100 days, your maximum 1-day loss will not exceed ₹18,450."

5. Smart Voice Phonetic Matcher

VOICE_SERVICE.PY

What It Does: Accurately identifies stock names from voice commands, even with heavy Indian accents, background noise, or voice typos.

How It Works: Compares spoken syllable patterns against 270 official NSE company names to find the closest match instantly.

Example: Spoken query "adacni entireprice" immediately matches ticker ADANI, opening the stock card and chart on screen hands-free.

Institutional Equity Modules (Overview, Dashboard, Domino & DNA)

WORKSPACES 01 – 04

Feature 1: Market Overview Radar & Live Order FlowMARKETOVERVIEWPAGE.JSX

Tracks 270+ liquid National Stock Exchange (NSE) market leaders cached in-memory. Evaluates live intraday Order Book Imbalances (buyer vs. seller queue) and institutional Volume-Weighted Average Price (VWAP) benchmarks to issue autonomous quantitative verdicts (*STRONG BUY, ACCUMULATE, HOLD, CAUTION*).

INSTITUTIONAL METRIC	HOW IT WORKS (PLAIN ENGLISH)	WHY IT PROTECTS THE TRADER
Order Book Imbalance (OBI)	Counts live buy orders waiting in the exchange queue vs. sell orders.	Reveals real buying pressure before prices move (+0.15 = buyers 15% stronger).
Anchored VWAP (AVWAP)	Calculates the true average price paid by all traders weighted by volume.	Acts as an institutional floor. If price is above VWAP, big institutions are in profit.
Historical Win Rate	Tests how many times this exact setup yielded a profitable trade in past years.	Replaces hopeful guesses with verified historical sample data (e.g., 68% wins over 2,589 setups).

Dalal Street Scenario: Reliance Industries trades at ₹1,285.20. MarketMind displays buyers at +10% strength, price above its ₹1,293.79 institutional benchmark, and calculates an upside target of ₹1,338.31 (+2.2%) with a clearly defined risk boundary.

Feature 2: Executive Multi-Workspace DashboardDASHBOARDPAGE.JSX

A professional command center inspired by institutional Bloomberg terminals. Aggregates four decoupled intelligence pods into a clean unified screen:

Workspace 01: Quant Matrix View: Ranks the entire 270-stock universe by institutional conviction, 1-day price speed, and sector momentum.

Workspace 02: Market Pulse Heatmap: Real-time visual treemap reflecting money rotation across 9 key Indian sectors (Banks, Metals, Energy, Auto, IT).

Workspace 03: Live Streaming News: Ingests official regulatory updates from SEBI adjudication circulars and RBI announcements with instant sentiment tags.

Workspace 04: Deep Telemetry Charts: Interactive TradingView canvas showing both candlestick patterns and smoothed trend curves with institutional bands.

Feature 3: Market Domino Predictor (Causal Macro Supply Chain)DOMINOPREDICTORPAGE.JSX

Stocks do not move in isolation. When global events occur (such as crude oil spikes or interest rate hikes), the Domino engine computes the ripple effect across connected company supply chains:

TRIGGER CATALYST	DIRECT FIRST EFFECT	CORPORATE MARGIN IMPACT	SMART SECTOR HEDGE
Crude Oil Rises +20%	Aviation fuel (ATF) becomes significantly costlier.	InterGlobe Aviation (IndiGo) airline profits fall (-210 bps).	ONGC profits expand (+₹1,120 Cr) as it sells oil at higher rates.
RBI Repo Rate Hikes +50 bps	Bank loan interest rates and borrowing costs increase.	Car and real estate buying slows down as EMIs go up.	Large commercial banks (HDFC Bank) earn higher lending margins.
USD/INR Depreciates +2%	Imported computer chips and electronics become costly.	Domestic electronic manufacturers see raw material costs rise.	IT exporters (TCS, Infosys) gain higher profits as they earn in US Dollars.

Feature 4: Stock DNA Fingerprint & Behavioral Pair TradingSTOCKDNAPAGE.JSX

Traditional sector grouping fails when large conglomerates operate across multiple industries. Stock DNA evaluates every company across 8 key financial traits (valuation, momentum, debt, company size, volatility, and trend slope):

Smart Stock Substitution: If an investor wants to profit from the booming Indian automotive sector but finds Tata Motors already overbought and expensive (+35% rally), Stock DNA compares financial fingerprints and suggests **Bharat Forge (89% similarity)** or **Ashok Leyland (84% similarity)** as identical, unheated alternatives at better valuations.

Risk Management, Chart Patterns & Alex Voice Copilot

WORKSPACES 05 – 08

Feature 5: Thesis Breaker (Zero-Hope Risk Architecture)

THEISISBREAKERPAGE.JSX

Retail traders frequently hold losing positions because of emotional denial ("hoping it will bounce back"). Thesis Breaker establishes a pre-calculated **Structural Invalidation Floor** anchored to major volume support:

Disciplined Exit Example: When buying Adani Enterprises at ₹2,950.00, the system sets an invalidation floor at ₹2,914.60. If the stock closes below this line, the buy thesis is proven wrong. MarketMind issues an immediate exit alert without emotional hesitation, saving the trader from catastrophic losses.

Feature 6: Candlestick Intel & Autonomous Pattern Detection

CANDLESTICKINTELPAGE.JSX

Scans historical candlestick bars to automatically identify high-conviction institutional accumulation patterns (Bullish Engulfing, Hammer Reversals, Support Reclaims). It eliminates retail trap breakouts by requiring **1.5x minimum volume confirmation**.

Pattern Detection Example: TCS breaks above its 20-day trading range with 2.4x average daily volume. The engine highlights a validated *Institutional Accumulation Formation* with 81% pattern confidence and an estimated 68% probability of continued upside.

Feature 7: Portfolio Simulator & 1,000-Scenario Risk Gauge

PORTFOLIOSIMULATORPAGE.JSX

Provides a risk-free virtual trading wallet (₹10 Lakh starting equity) to practice institutional trade setups with realistic market conditions. Daily portfolio tail-risk is checked using **1,000 randomized market stress simulations**:

Clear Risk Certification: Instead of confusing equations, the engine gives a straightforward risk statement: *"Across 95 out of 100 normal trading days, your maximum 1-day portfolio drawdown will not exceed ₹18,450 (1.85%)."*

Feature 8: Alex Copilot — Autonomous Financial Voice Assistant

USEVOICEAGENT.JS & VOICE_SERVICE.PY

Alex is an intelligent voice copilot designed specifically for Indian stock markets. It listens continuously and translates speech directly into active UI screen controls:

VOICE FEATURE	HOW IT WORKS (PLAIN ENGLISH)	WHY IT WAS BUILT
Word-by-Word Streaming STT	Captures words in real-time as the user speaks, showing live text immediately.	Gives instant visual feedback so the user knows the AI is actively listening.
2.2s Natural Pause Detection	Waits 2.2 seconds of silence before processing spoken commands.	Allows users to pause and think without being rudely cut off mid-sentence.
Multi-Turn Memory Stack	Remembers the active stock in conversation.	Enables natural follow-up questions: <i>"What is its target?"</i> remembers Reliance.
Acoustic Echo Cancellation	Detects when Alex is speaking and suppresses microphone feedback.	Prevents laptop speakers from triggering an endless loop of self-talking.
Dual Voice Speech Synthesis	Deepgram realistic AI voice + browser fallback synthesizer.	Delivers natural human-like voice with 100% reliability even during slow internet.

Autonomous Screen Navigation: The user says: *"Show me adacni entirerpice"*. Alex corrects the typo to ADANIENT, switches the screen to Market Overview, scrolls the card to center, expands the chart, and speaks: *"Adani Enterprises trades at ₹2,950.00 (+0.41%), target ₹3,003.10 with stop-loss at ₹2,914.60."*

End-to-End Technology Stack & Cross-Screen Action Matrix

FULL-STACK ARCHITECTURE

Production Technology Stack Rationale		
LAYER	CHOSEN TECHNOLOGY	WHY WE SELECTED IT
Backend Framework	FastAPI (Python 3.11+)	Ultra-fast asynchronous API with native support for Python numerical math libraries (NumPy, SciPy).
Frontend Framework	React 18 + Vite	Blazing-fast build times (460ms) with instant Hot Module Replacement for smooth trader UI.
Styling Architecture	Vanilla CSS (Design Tokens)	Zero runtime CSS bloat. Clean institutional light/dark theme design inspired by Bloomberg terminals.
Voice Recognition	Web Speech + Deepgram Nova-2	Sub-second speech-to-text accuracy tuned for Indian English and regional accents.
AI Speech Synthesizer	Google Gemini 2.5 Flash-Lite	Lightning-fast AI responses (< 1.8s) translating complex financial metrics into crisp 35-word voice answers.
Stock Data Feed	Yahoo Finance + In-Memory Cache	Instant data retrieval (0.0001ms) from memory cache for 270+ Indian equities.

Cross-Screen Voice Synchronization Matrix		
Alex Copilot dispatches voice action events that automatically control and navigate the entire application:		
SPOKEN VOICE COMMAND	TARGET SCREEN	AUTONOMOUS UI ACTION EXECUTED
"Show Reliance Industries"	Market Overview	Finds RELIANCE, scrolls card to center screen, and expands live price chart.
"What if crude rises 20%?"	Domino Predictor	Navigates to Domino screen, sets crude oil slider to +20%, and shows supply chain impact.
"Find twin for Tata Motors"	Stock DNA	Opens Stock DNA and matches Bharat Forge (89% twin match) on the radar screen.
"Stress test HDFC Bank thesis"	Thesis Breaker	Opens Thesis Breaker and highlights the calculated invalidation exit line in red.
"Show Syрма SGS candles"	Candlestick Intel	Switches to Candlestick Intel, loads OHLCV chart, and highlights accumulation patterns.
"Calculate portfolio risk"	Portfolio Simulator	Navigates to Portfolio screen, runs 1,000 market simulations, and updates VaR risk gauge.
"Show latest news from SEBI"	Latest News	Navigates to News screen and filters official SEBI regulatory circulars.
"Open Sector Intelligence"	Sector Intelligence	Loads 9-sector heatmap showing where institutional money is moving today.
"Open Market Pulse"	Executive Dashboard	Switches Dashboard view to the Market Pulse real-time visual treemap.

3-Tier Resilient Fallback Architecture (Zero Freeze Guarantee)		
Financial applications must never freeze or crash during live market hours. MarketMind AI enforces a 3-layer safety net:		
Tier 1: Live Market Feed: Queries live exchange endpoints for real-time 5-minute intraday stock candles.	Tier 2: 2.5s Timeout Guard: If an external internet call takes longer than 2.5 seconds, the request is safely cancelled.	Tier 3: In-Memory Fallback: Renders verified historical closing curves so charts and UI never get stuck on loading spinners.

Evaluator Viva Defense: Top 7 Technical & Conceptual Questions

INTERVIEW Q&A

Q1: "Why did you use explainable quantitative logic instead of a Deep Learning Neural Network (LSTM)?"

Defense: Deep learning neural networks are "black boxes" that overfit to market noise and hallucinate unpredictable outputs. Institutional hedge funds like Citadel and Renaissance Technologies do not let black-box neural networks make live trading decisions because their risks cannot be audited. Our 5 quantitative engines provide 100% explainable, deterministic logic with sub-5 millisecond response times and zero hallucination risk.

Q2: "What is the exact role of Gemini AI in MarketMind, and what happens if Gemini is offline?"

Defense: Gemini does not calculate numbers, stock targets, or risk percentages—all math is computed by our fast Python backend. Gemini's role is an *Institutional Voice Synthesizer*: it takes our calculated numbers and formats them into a polite, concise 35-word spoken response. If Gemini goes offline or lags past 2.2 seconds, our backend instantly uses an offline fallback template that speaks the exact price, target, and stop loss without delay.

Q3: "How does the voice system understand misspelled stock names or Indian accents?"

Defense: It removes conversational filler words (*"show me", "share price"*), breaks the audio into phonetic pieces, and compares the character sequence against 270 official company names and common aliases. If similarity is above 80%, it immediately matches the stock (e.g., *"adacni entirerpice"* → *ADANI ENT*).

Q4: "How does Acoustic Echo Cancellation prevent Alex from talking to himself through the mic?"

Defense: When Alex speaks through the laptop speakers, the mic could pick up his own voice and treat it as a new user command. We track an active audio flag (*isPlayingAudioRef*). If incoming sound matches what Alex is currently speaking, the audio input is blocked. We also enforce a 450ms silence pause after speaking before re-opening the mic.

Q5: "How does the platform handle multi-turn conversations (pronouns like 'it')?"

Defense: The client maintains a rolling memory of recent conversation turns. If a user first asks *"How is Reliance doing?"* and then follows up with *"What is its stop loss?"*, the backend checks the history stack, identifies that RELIANCE was the active subject, and answers for Reliance without forcing the user to repeat the company name.

Q6: "How does MarketMind handle post-market hours (after 3:30 PM IST)?"

Defense: The platform tracks official Indian Market Hours (09:15 to 15:30 IST, Monday through Friday). After 3:30 PM, it automatically switches to *POST-MARKET / CLOSED* status, freezes prices to official end-of-day settlement rates, and displays preparation analytics for the next morning's market opening.

Q7: "What regulatory safety guardrails prevent reckless financial advice?"

Defense: If a user asks speculative questions like *"How can I double my money in 1 week?"* or *"Should I take a loan to trade?"*, strict SEBI prudence rules intercept the query. Alex politely refuses, warns against borrowed capital, and clarifies that MarketMind is an educational quantitative platform with zero guaranteed profits.

Production Performance & Audit Certification

Build Speed: 460ms (Vite 8.2 Production Bundle)

Bundle Footprint: 339 KB CSS | 361 KB Gzipped JS

In-Memory Universe: 270+ Liquid NSE Equities